Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident

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Summary

Data Scientist with a strong background in **AI Research**, with experience across **healthcare**, **NLP**, and **geoscience** domains. Builds **end-to-end ML pipelines**, conducts **statistical and spatial analysis**, and deploys models that hold up in production. Develops **agentic AI pipelines** using **RAG**, **LLMs**, and **MCP**. Published **peer-reviewed research** across **IEEE**, **SAGE**, and **Springer Nature**. Comfortable working across the full stack from raw data to results, and collaborating directly with domain experts. Proficient in **Python**, **SQL**, **R**, **PyTorch**, and **TensorFlow**.

Education

The City College of New York — New York, NY
M.S. in Data Science and Engineering | GPA: 3.83/4.0 — *Dec 2027*

East West University — Dhaka, Bangladesh
B.S. in Computer Science and Engineering | GPA: 3.74/4.0 — *Dec 2023*

Technical Skills

Languages: Python, SQL, R
Data Analysis: NumPy, Pandas, SciPy, Excel
Data Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau
AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK, Word2Vec
LLMs & Agents: LLMs, RAG, MCP, Hugging Face, ChromaDB
Database & Big Data: MySQL, MongoDB, Snowflake
MLOps & Cloud: AWS, MLflow, Git/GitHub, Docker, Vercel

Work Experience

Data Analyst — Oct 2025 - Feb 2026, Jul 2026 - Present
The City College of New York — *New York, NY*

- Analyze **institutional data** using **Excel, Tableau,** and **Python**, surfacing patterns that shape **advising and student engagement programs**.
- Conduct **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach.
- Build **analytical reports** and **dashboards** that surface trends and anomalies across departments, replacing gut-feel decisions with **data-backed ones**.

Research Assistant — Dec 2025 - Present
CUNY Research Foundation — *New York, NY*

- Develop an **agentic AI pipeline** using **RAG**, **LLMs**, and **MCP-connected tools** for natural language querying over complex geoscience databases, as part of an **NSF-funded** geoinformatics project.
- Bridge **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specs and iterating on pipeline outputs for interpretability and scientific relevance.
- Design the **end-to-end technical stack**: **vector database architecture**, domain-specific **embedding models**, **LLM optimization**, and **cloud deployment** for a production-ready geoscience research tool.

Data Evangelist Intern — Feb 2026 - May 2026
DataCamp — *New York, NY*

- Designed and maintained **scalable data pipelines** to collect, integrate, and validate **multi-source data** through automated scripts, APIs, and web scraping.
- Built **data applications** and **analytical workflows** powered by **LLMs**, integrating **OpenAI** and **Hugging Face** models into user-facing tools for exploring key metrics.
- Analyzed data to surface trends and key drivers, packaging findings into **visualizations** and structured reports.

Research Experience

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- **Achieved 97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- **Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

Papers Under Review

Multimodal Learning in Medical Imaging: Methods, Benchmarks, Evaluation, & Clinical Translation – *Elsevier*

- Conducted a structured review of **multimodal learning in medical imaging**, proposing a unified taxonomy while identifying key gaps in **robustness, reproducibility, and benchmark standardization** across healthcare AI systems.

Data Projects

NYC HIV/AIDS Trend Analysis & Predictive Modeling | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered **multi-year NYC Open Data** on HIV/AIDS diagnoses into analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis** across NYC, uncovering persistent geographic and demographic disparities.
- Applied **Random Forest models ($R^2 = 0.65$)** to identify high-impact predictors for public health targeting.

Global Population Growth Modeling & Forecasting | *R, Tidyverse, ggplot2, leaps, caret*

- Engineered a **longitudinal time-series dataset** from World Bank indicators across **200+ countries** and **60+ years** of demographic trends.
- Conducted multi-decade demographic analysis by region and income group, revealing **divergent growth trajectories** between developing and developed economies.
- Developed forecasting models achieving $\mathbf{R^2 = 0.755}$ using **rolling-window cross-validation**.

NYC Restaurant Inspections Case Study | *Python, Pandas, NumPy, Plotly, JavaScript*

- Built a reproducible **ETL pipeline**, collapsing **83,000+** violation-level records into an analysis-ready inspection dataset.
- Applied **descriptive and discontinuity analysis** to inspection scores, quantifying closure risk across violation categories.
- Designed an interactive **Plotly-based visualization layer** to make complex statistical patterns interpretable for a non-technical audience.

AI/ML Projects

Deep Learning Based Wound Classification | *Python, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib*

- Improved validation accuracy from **89.8% to 98.5%** by redesigning the CNN architecture with added convolutional layers, batch normalization, dropout, and L2 regularization.
- Applied **oversampling** to correct significant class imbalance across 10 wound categories, improving fairness and reliability.
- Reduced loss to **0.1061** using EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks to control training.

Intrusion Detection System | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- **Processed and analyzed 2.5+ million network traffic records** through EDA, uncovering key patterns in anomaly distribution.
- Reduced memory usage by **47.5%** through numerical datatype downcasting.
- **Achieved 97% median accuracy** detecting network anomalies using machine learning models.